UE26CY151A Engineering Chemistry · Unit 2 · Electrochemistry and Corrosion

# Flowcharts and visuals

Procedures, decisions, concept maps and comparison tables.

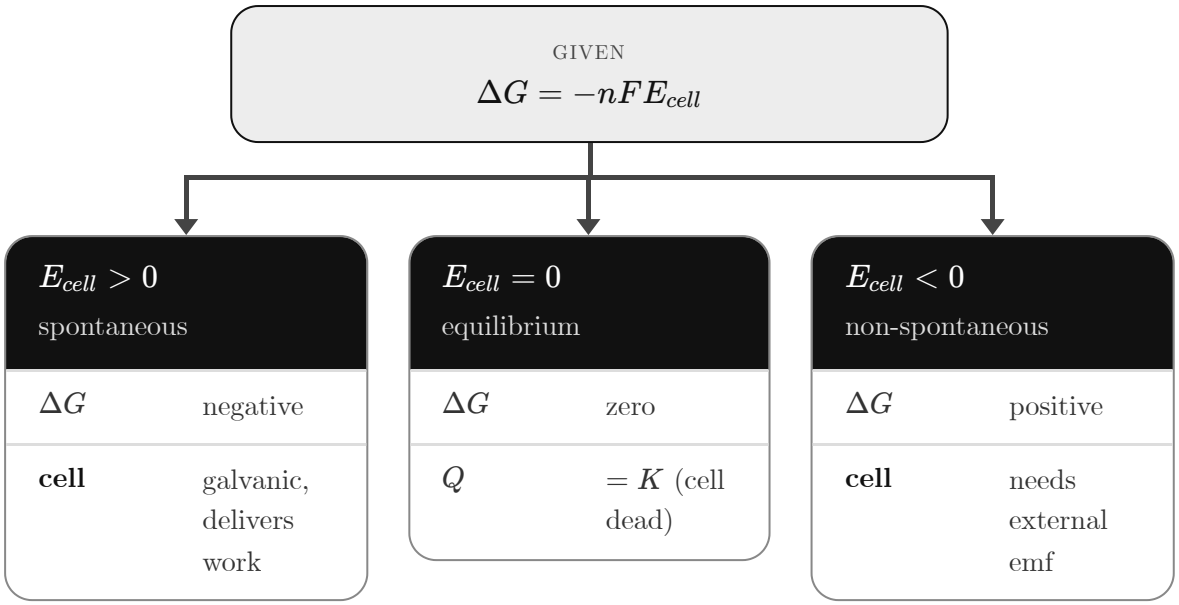
Class 002 · Nernst equation and its derivation · compiled 17 October 2026

| | |
|---|---|
| ESA ★★★ | Asked repeatedly in past ESA papers |
| ESA ★★ | Asked in past ESA papers or computed in class-work problems |
| ESA ★ | Conceptual: taught in class, not examined so far |
| ADDED FOR CLARITY | Standard context that is not in the class material |
| TRANSCRIPT ONLY | Said in the lecture but not on any slide |
| VIDEO ONLY | Seen only in a frame of the class video |
| NOT ON CURRENT SLIDES | From older material or past papers, not this year's slides |
| S002·4 | Deck 002, slide 4 (rendered .pptx) |
| T001 00:21:42 | Class 001 AV-summary transcript (covers classes 1 and 2), segment start |
| PDF p.13 | Consolidated Unit 2 slide PDF (older syllabus), page 13 |
| ESA Jan-25 s1 2a | Past paper: Jan 2025, set 1, question 2.a |

## Contents

1. Figure F2.1 · Derivation of the Nernst equation, step by step.
2. Figure F3.1 · Which form of the Nernst equation?
3. Figure F4.1 · Solving a cell-emf numerical with the Nernst equation.
4. Figure F1.1 · What the sign of $E$ says about $\Delta G$.
5. Nernst equation: single electrode vs cell
6. Signs of $\Delta G$, $E$ and $K$

# 1 Free energy and electrode potential




***Remark.*** The middle branch ($E = 0$ at equilibrium) is added for clarity; the lecture gives the two outer cases.

**Figure F1.1.** What the sign of $E$ says about $\Delta G$. ESA ★★

T001 00:16:39 · T001 00:16:51 · PDF p.10

### Signs of $\Delta G$, $E$ and $K$

| Reaction | $\Delta G$ | $E_{cell}$ | $K$ |
|---|---|---|---|
| **Spontaneous** | $< 0$ | $> 0$ | $> 1$ |
| **At equilibrium** | $0$ | $0$ | $Q = K$ |
| **Non-spontaneous** | $> 0$ | $< 0$ | $< 1$ |

ADDED FOR CLARITY T001 00:16:39 · S002·3

# 2 Derivation of the Nernst equation for a single electrode

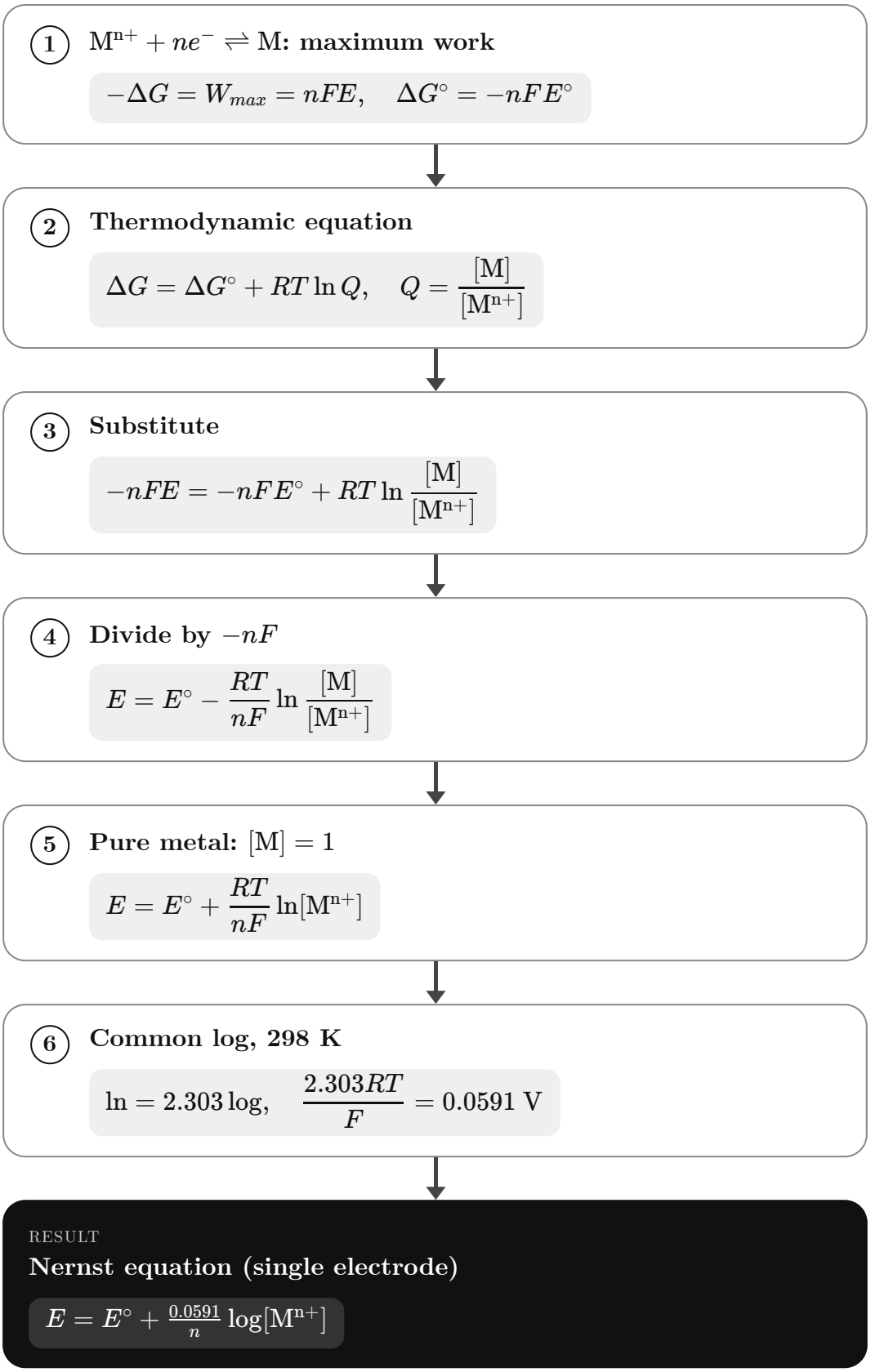
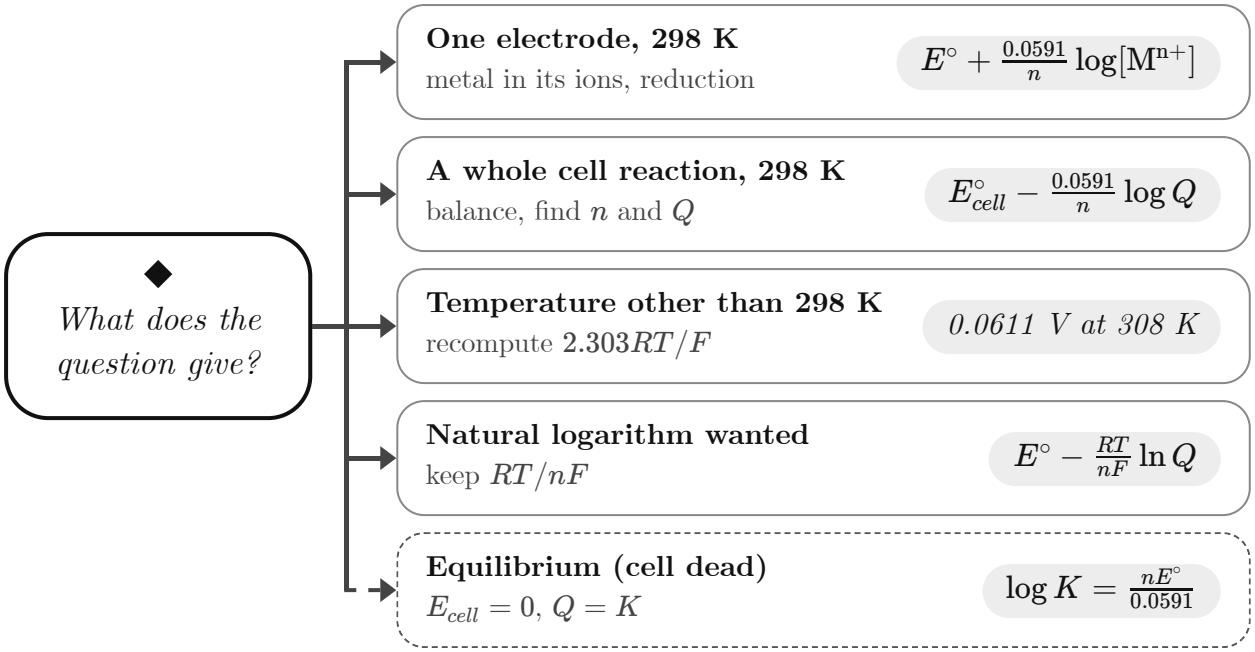



**Figure F2.1.** Derivation of the Nernst equation, step by step. ESA ★★★

S002·3 · S002·4 · S002·5 · T001 00:19:38 · T001 00:23:50 · ESA Dec-24 2a

# 3 Using the single-electrode equation: concentration and temperature

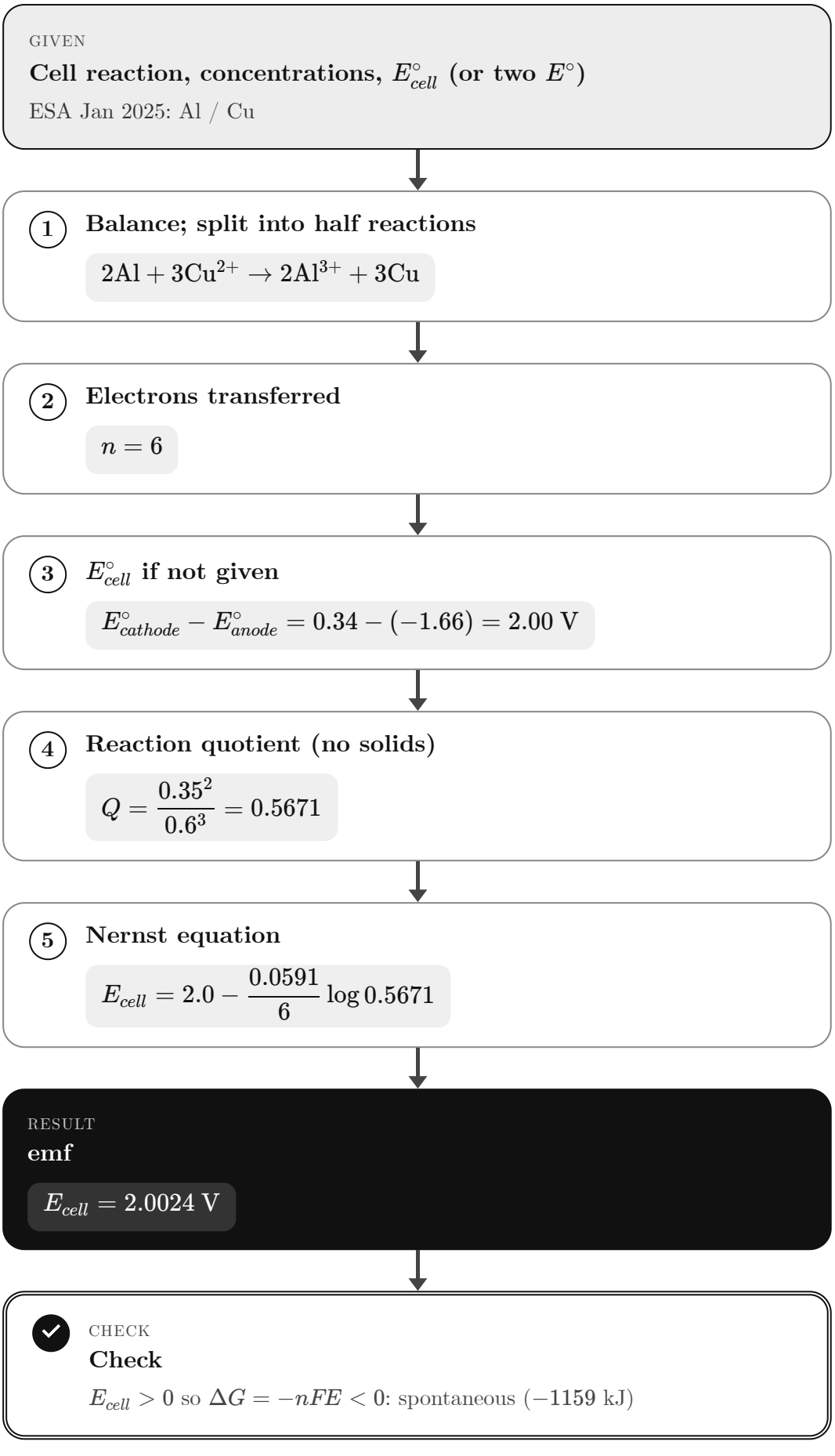



***Remark.*** The dashed route (equilibrium constant) is added for clarity; it was not taught in class 002.

**Figure F3.1.** Which form of the Nernst equation? ESA ★★

S002·4 · S002·5

# 4 Nernst equation for a cell



***Remark.*** In a 6-mark question write the derivation (F2.1, cell form) before the calculation.

***Exam trap.*** Balance first: the paper prints "+ Cu". Wrong $n$ gives the wrong emf.

**Figure F4.1.** Solving a cell-emf numerical with the Nernst equation. ESA ★★★

S002·5 · ESA Jan-25 s1 2a

### Nernst equation: single electrode vs cell

| | Single electrode | Cell |
|---|---|---|
| **Reaction** | $\mathrm{M^{n+}} + ne^- \rightleftharpoons \mathrm{M}$ | $a\mathrm{A} + b\mathrm{B} \rightleftharpoons c\mathrm{C} + d\mathrm{D}$ |
| $Q$ | $[\mathrm{M}]/[\mathrm{M^{n+}}] = 1/[\mathrm{M^{n+}}]$ | $[\mathrm{C}]^c[\mathrm{D}]^d/[\mathrm{A}]^a[\mathrm{B}]^b$ |
| $E$ **means** | reduction (electrode) potential | emf $E_{cell}$ |
| **Standard value** | $E^\circ$ | $E^\circ_{cell} = E^\circ_{cathode} - E^\circ_{anode}$ |
| **Form at 298 K** | $E^\circ + \frac{0.0591}{n}\log[\mathrm{M^{n+}}]$ | $E^\circ_{cell} - \frac{0.0591}{n}\log Q$ |
| $n$ | electrons in the half reaction | electrons in the balanced overall reaction |
| **ESA** | Dec 2024 Q2.a (5 marks) | Jan 2025 Q2.a (6 marks) |

S002·4 · S002·5